Project Proposal for Battery Management System (BMS)

Sponsored by Viking Motorsports (VMS)

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ECE Capstone
Senior Project Development

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1 Executive Summary

Our team will design the next-generation battery system for PSU's electric Formula race car, EV27, in partnership with VMS and our faculty advisor. The current EV program does not have a usable accumulator, so the car won't run. Over the next two terms we will create a complete, competition-ready design for the high voltage battery pack and its BMS, plus working low voltage prototype that demonstrates the core functions. The car we're designing this battery for will compete in a Formula-style racing competition where universities bring their own cars to do a race competition.

2 Background and Research

VMS is our sponsor. They are a medium-sized student group that participates in the Formula SAE Electric event.

Electric vehicles rely entirely on large battery packs as their primary energy source. These battery packs are made up of many individual cells, typically lithium-ion, that must operate within some strict electrical and thermal limits and requirements in order for them to remain safe, efficient, and long-lasting. Due to the fact that lithium-ion battery chemistry is sensitive to properties such as overcharging, over-discharging, overheating/temperature, and cell imbalance, electric vehicles often require a dedicated subsystem called the BMS, to monitor and protect the battery.

These systems exist because lithium-ion battery packs can be extremely dangerous if not carefully monitored and diagnosed. The individual cells can be damaged due to things such as overcharging, over-discharging, and overheating. So, electric vehicles incorporate a battery management system to act as a protective mechanism using a combination of sensing, diagnostics, and protective circuit logic within the system. The system ensures that the battery operates only within safe electrical and thermal limits, which can in turn improve safety, extend battery life, and optimize performance.

The battery management system continuously measures key parameters such as cell voltages, current flow, pack and cell temperatures, and the estimated state of charge. It then processes the data to compute additional metrics, such as pack health, state of energy, imbalance levels, and thermal stability. In addition to monitoring, the BMS controls balancing circuitry using techniques such as active or passive cell balancing, to keep individual cells at consistent charge levels. Additionally, the BMS manages protective actions such as overvoltage, undervoltage, and overcurrent cutoffs. Also, the BMS communicates diagnostic information, warnings, and operational status to the vehicle's electronic control unit.

Within the larger electric vehicle system, the BMS serves as the central system responsible for determining whether the high-voltage battery pack is in a safe condition to deliver power to the vehicle. For instance, the electronic control unit depends on the BMS for accurate state of charge information, temperature status, and di-

agnostic reports. The motor, motor controller, charger, and thermal management system all rely on BMS outputs to function properly. Therefore, the BMS is not an isolated system but rather a core subsystem and safety component that interfaces with a lot of other components in the vehicle.

In practice, a battery management system is an electronic system that manages a rechargeable battery pack. Its operation follows a continuous loop of measure, protect, balance, and inform. First, it measures the fundamental state of the battery by using a network of sensors and integrated circuits to read the voltage of every individual cell, the total current flowing in or out of the pack, and its temperature at key locations. Second, it uses this data to protect the battery from damage by comparing the readings against safe limits. For instance, if it detects a dangerous condition like overvoltage, undervoltage, overcurrent, or extreme temperatures, it will command a safe shutdown by disconnecting the pack. Third, to maximize the pack's capacity and longevity, it balances the cells, using techniques such as active or passive cell balancing to bleed off excess charge from higher voltage cells, ensuring all cells reach a uniform state of charge. Finally, it informs the larger system by processing the raw measurements into useful information such as state of charge and health, and it communicates this data, along with any warnings, to other devices.

According to the project description provided by Viking Motorsports, the electrical team has completed early conceptual research for the Formula SAE Electric platform. This foundational work includes preliminary investigation into CAN bus communication for vehicle subsystem networking, initial evaluation of cell monitoring integrated circuits and measurement topologies for per-cell voltage sensing, and high-level planning of the system-level safety architecture required to meet Formula SAE Electric fail-safe shutdown and interlock rules. These efforts established an initial understanding of the technical challenges and constraints associated with a high-voltage accumulator system but did not result in an implemented or tested BMS design.

At the start of this capstone project, no functional BMS hardware or firmware prototype exists, and there is no previous generation of a Viking Motorsports BMS to revise or extend. As a result, this project represents a ground-up design and development effort rather than an incremental revision. Our team is responsible for

transforming the sponsor's early conceptual research into a testable BMS prototype that performs real-time measurement, protection, balancing, and diagnostic functions that are suitable for future vehicle integration.

If nothing has been done on this project yet, are there comparable products or existing alternatives? Be creative here, what are similar technologies or direct competitors? Include them if they're relevant or help describe your project.

See commercial off the shelf products and open source projects sections of the project proposal.

2.1 Research

Final Summary: The BMS Knowledge Framework Matrix

This table represents the complete structure of our Proposal.

Layer	Module	Key Technologies / Keywords	Role
1. Sensing	**Voltage/Temp**	AFE Chips, NTC Thermistors, RC Filters	EE + ME
	Current	Hall Effect Sensor	EE + ME
2. Logic	**State Estimation**	SOC (Coulomb Counting), SOP (Derating)	CE
	Protection	OVP, UVP, OTP, OCP (Hard Limits)	CE
3. Acting	**Balancing**	**Passive Balancing** (Bleed Resistors)	EE
	HV Control	AIRs (Contactors), Pre-charge Circuit	EE + ME
4. Safety	**Isolation**	HV/LV Galvanic Isolation, Spacing	ME + EE
	Interlock	Safety Loop (Hardware series loop)	ME + EE
5. Comms	**External**	CAN Bus (Isolated), DBC File	CE
6. Env	**Integrity**	EMI/EMC, Shielded Cabling, Thermal Mgmt	ME + EE

2.1.1 Commercial Off-the-Shelf Products

Here are some options available to us for this project:

1. **Orion BMS 2:** The Orion BMS 2 is a widely adopted **Centralized** solution in the FSAE and EV conversion sectors. While it integrates all measurement

and control units into a single enclosure, offering mature reliability and EMI immunity, its architecture requires routing extensive sensing harnesses from all points of the battery pack to a central controller. This results in complex wiring and increased system weight. This serves as a critical counter-reference for our project: while centralized designs offer EMI advantages, their disadvantages in harness management within a high-voltage (300V+) and weight-critical racing environment validate our decision to pursue a Distributed Architecture.

2. **Stafl Systems BMS1000M:** Stafl Systems utilizes a **Master-Slave Distributed Topology**. The system consists of a master controller (BMS1000M) and multiple satellite monitoring modules (BMS1101S), utilizing differential signaling (e.g., IsoSPI) for communication. This architecture significantly reduces the length of analog sensing wires, effectively lowering wiring complexity within the high-voltage pack. As a direct industry benchmark, Stafl demonstrates the engineering feasibility of distributed architectures for segmented packs. However, its Closed-Source nature limits secondary development for specific research algorithms, restricting its role to a hardware architecture reference.

3. **EMUS G1 BMS:** EMUS G1 demonstrates a highly **Modular** distributed strategy, supporting the connection of multiple independent battery modules via CAN bus. This product employs robust galvanic isolation strategies and communication protocols ideal for physically separated battery enclosures (e.g., side-pod layouts). Given the potential requirement to split our 400V pack into multiple physical modules, EMUS provides a critical technical reference for High-Voltage Galvanic Isolation and data synchronization in our safety loop design.

2.1.2 Open Source Projects

Here are some projects we can borrow from:

1. foxBMS 2: foxBMS 2 is a comprehensive, professionally developed open-source BMS platform from the Fraunhofer research institute, explicitly designed as

a research and development platform for high-performance, high-availability battery systems. Its architecture is a master-slave distributed topology, where a central BMS-Master (ARM Cortex-R5) communicates via an interface board with daisy-chained Slave boards that handle cell monitoring (voltage/temperature) and passive balancing. The primary benefit is its exceptional depth. For instance, it offers a complete, well-documented hardware and software stack, including a built-in interlock line for safety shutdowns, CAN bus communication for system integration, and a flexible design that supports from a few cells up to several hundred kWh. The permissive licenses allow full use and modification. The notable drawback is its scale and complexity. For instance, as a generic research platform, it is not optimized for the strict space, weight, and FSAE-rule constraints of a race car. Its default use case is stationary storage, requiring significant adaptation for a mobile, high-vibration environment such as a race car. Nonetheless, it still serves as a great resource for robust system architecture, safety implementations such as interlock monitoring, and professional-grade, modular firmware structure.

2. Libre Solar BMS: The Libre Solar BMS is a compact, fully integrated open-source BMS designed for stationary and small-scale mobile DC energy systems (12V-48V, up to 16 cells). Its architecture is highly centralized, combining the BMS Analog Front-End (Texas Instruments BQ76952), microcontroller (ESP32-C3), power MOSFETs, and communication interfaces (CAN, UART, I2C, USB, Bluetooth) on a single PCB. The primary benefit is its production-ready, all-in-one design that demonstrates efficient integration of core BMS functions such as cell monitoring, protection, passive balancing, and contactor control, within a minimal footprint. Its firmware, built on the Zephyr RTOS, provides a professional reference for implementing safety logic, the ThingSet protocol for configuration, and state-of-charge estimation. The notable drawback is its scale. For instance, it is designed for 60V systems with currents up to 100A, its cell count and power handling are far below the requirements of a 400V+ FSAE powertrain. Its centralized architecture is also less ideal for the long, distributed battery pack typical in a race car frame. Nonetheless,

it still serves as an excellent reference for PCB-level integration, component selection, and a clean, modular firmware structure that can inform the design of sub-modules within our larger, distributed system.

3. Green BMS: Green BMS is a hobbyist-developed, smart BMS designed for small-scale lithium battery packs such as for electric scooters or solar storage. Its architecture is a modular distributed system, using simple Cell Modules based on ATtiny microcontrollers to measure individual cell voltage/temperature, which report via I2C to a central Control Unit (Arduino Mega). The system includes a separate "Limiter" unit for contactor control and an Android app for monitoring. The primary benefit is its exceptional transparency and simplicity, it demonstrates the absolute core functions of a BMS (monitoring, protection, balancing) with minimal complexity, using widely understood components (Arduino, ATtiny). This makes it an excellent educational tool for understanding the fundamental data flow and logic of a BMS before engaging with more complex, integrated circuits. The significant drawbacks are its lack of automotive robustness, missing critical safety features (high-voltage isolation, CAN bus, fail-safe hardware interlocks), and limited scalability (I2C communication is unsuitable for long daisy-chains in a large pack). Nonetheless, it still serves as a valuable conceptual reference for system partitioning and basic

2.1.3 Papers

Here is some extra papers we read during our research:

1. Evaluation and Comparison of Battery Cell Balancing Methods: This paper provides a structured, quantitative framework for selecting a cell balancing topology, directly addressing a key design decision. It categorizes and compares the three main balancing methods, dissipative (passive), capacitive energy transfer (active), and runtime balancing (active with converters), against critical metrics such as balancing speed, efficiency, complexity, and cost. This comparison is very valuable for our requirement to "support both active and passive cell balancing." It provides an objective basis to decide whether to

implement only passive (simpler, cheaper) or also include an active method (faster, more efficient) in our prototype.

2. Review of Battery Cell Balancing Methodologies for Optimizing Battery Pack Performance in Electric Vehicles: This paper provides a critical system-level perspective on cell balancing for batteries in electric vehicles, advocating for hybrid balancing methods that combine passive and active balancing techniques to leverage their respective advantages. This directly aligns with our requirement to "support both active and passive cell balancing," suggesting an approach where both techniques work together in a complimentary manner (passive for continuous maintenance, active for rapid pre-charge equalization). Additionally, it highlights environmental factors, specifically operating temperature, thermal uniformity, and mechanical vibration, as crucial but under-researched variables that affect long-term balancing performance and cell degradation.

3. State-of-the-Art and Energy Management System of Lithium-Ion Batteries in Electric Vehicle Applications: Issues and Recommendations: This paper provides a holistic overview of Li-ion battery technology and the complete scope of a Battery Management System (BMS). It systematically covers core BMS functions critical to our project, such as cell condition monitoring, state estimation (SOC/SOH), charge/discharge control, protection circuitry, cell equalization, and thermal management. It is particularly valuable for understanding the interdependencies between different BMS subsystems and for identifying key challenges, such as managing temperature gradients and diagnosing faults, which are actively relevant to the harsh, high-vibration environment of a Formula SAE vehicle.

3 Product Design Specification

3.1 Product Overview

This paper provides a holistic overview of Li-ion battery technology and the complete scope of a BMS. It systematically covers core BMS functions critical to our project, such as cell condition monitoring, state estimation (SOC/SOH), charge/discharge control, protection circuitry, cell equalization, and thermal management. It is particularly valuable for understanding the interdependencies between different BMS subsystems and for identifying key challenges, such as managing temperature gradients and diagnosing faults, which are actively relevant to the harsh, high-vibration environment of a Formula SAE vehicle.

3.2 Stackholders

Our stakeholders include everyone who is directly or indirectly affected by the accumulator and BMS system. At the top level, we have our industry and university sponsors, who provide funding, facilities, and oversight, and care about safety, and reliability. The vehicle team are key users of our design: they integrate the pack into frame, rely on our interfaces, and depend on our documentation to build and service the system. Future student teams who will build, maintain, and possibly upgrade the pack are another important group, because our documentation will either make their work easier or harder.

3.3 Requirements

Must:

- Continuously monitor all cell voltages and pack temperature.
- Provide reliable overvoltage, undervoltage, and overcurrent protection.
- Support both active and passive cell balancing.
- Communicate with the ECU over the CAN bus.

- Operate within SAE EV voltage limits and isolation standards.
- Include a fail-safe shutdown or interlock function for critical faults.

Should:

- Include onboard data logging capability.
- Use modular architecture to scale for different battery pack sizes.
- Support self-diagnostic startup and fault recovery routines.

May:

- Include an integrated display or diagnostic interface for local readings.
- Support wireless telemetry or USB debugging for testing.

3.4 Initial Product Design

3.4.1 Hardware Architecture Block Diagram

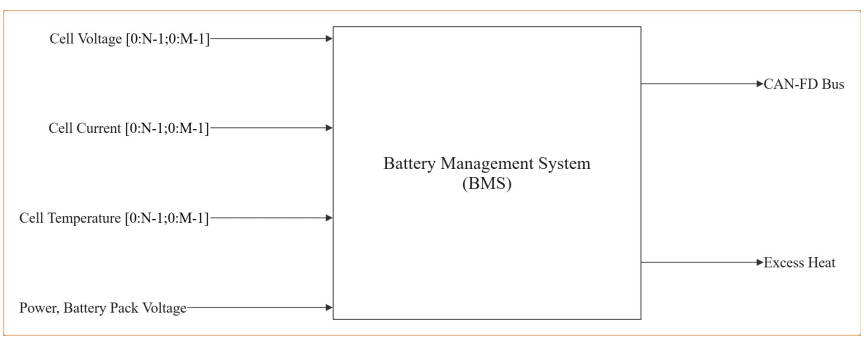


Figure 1: L0 Diagram

Module	Battery Management System
Inputs	Cell Voltage [0:N-1;0:M-1], Cell Current [0:N-1;0:M-1], Cell Temperature [0:N-1;0:M-1], and Power, Battery Pack Voltage
Outputs	CAN-FD Bus, Shutdown Circuit (SC) Signal, and Excess Heat
Functionality	Monitors for critical faults, reports measurement & faults to the ECU, and balances charge across the battery.

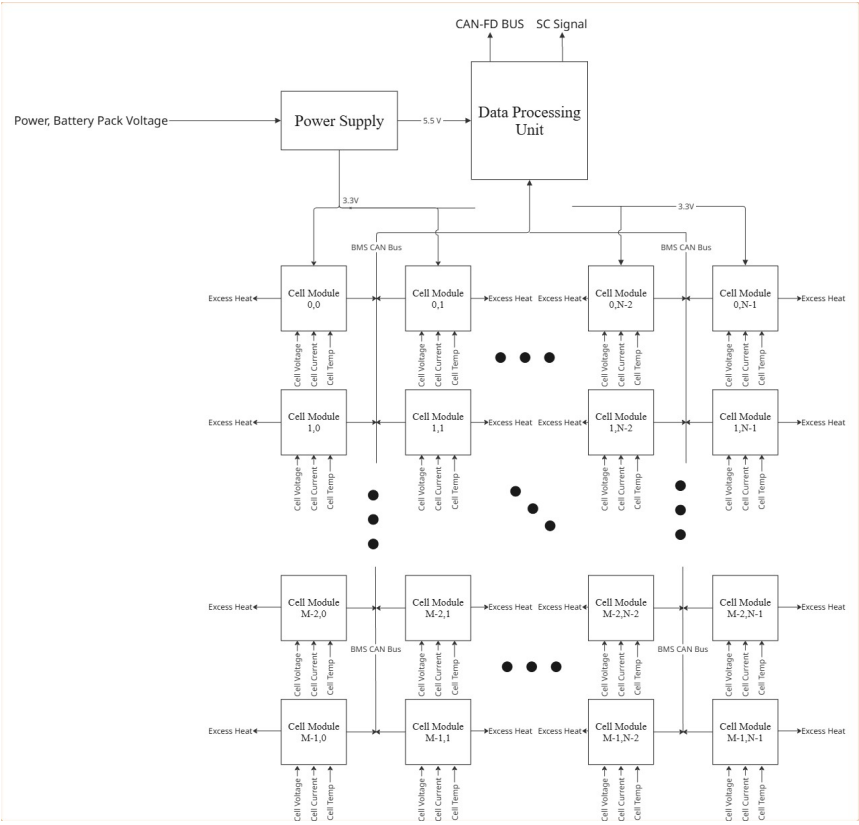
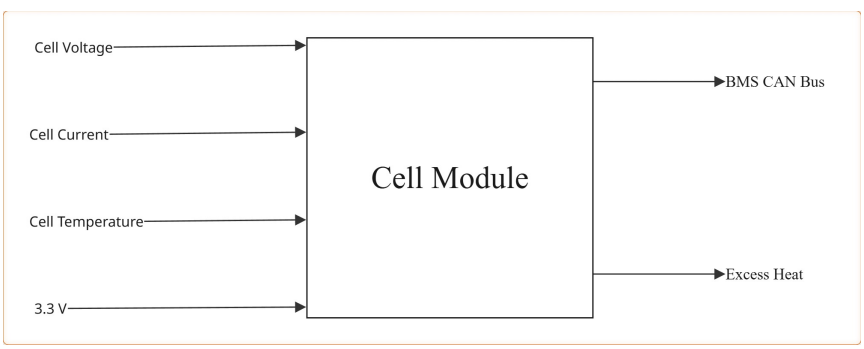
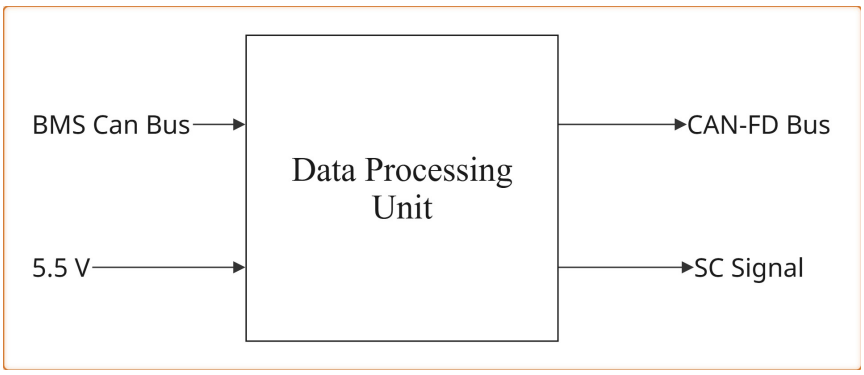


Figure 2: L1 Diagram



Module	Cell Module
Inputs	Cell Voltage, Cell Current, Cell Temperature, and 3.3V Power
Outputs	BMS CAN Bus and Excess Heat
Functionality	Measure the Voltage, Current and Temperature and convert them into digital data with the specific cell number that is sent to the Data Processing Unit. A resistor in the module passively balances the cell's voltage to be constant by dissipating excess charge as heat.



Module	Data Processing Unit
Inputs	BMS Can Bus and 5.5V Power
Outputs	CAN-FD Bus
Functionality	Collects all of the measurements from the Cell Modules and formats the data into a format accepted by the ECU to be sent over the CAN-FD Bus. If any major fault like Overvoltage is detected it will open the shutdown circuit to turn off the vehicle.

3.4.2 Software Architecture

This system will be programmed in Embedded C. The following Data flow table describes how the program will function.

Module	Inputs	Outputs
Measurement & Filtering	Raw cell voltages, pack voltage, pack current, temperature sensor readings	Filtered cell voltages, pack current, pack temperature
Balancing controller	Cell voltages, balancing thresholds, fault status	Balancing enable signals/ duty\ to each monitored cell or group
CAN-FD communication	Measurements, SOC/SOH, Fault status, config data	CAN-FD frames to ECU (telemetry, warnings, faults), received config/commands from ECU
Fault Detection and Protection Logic	Filtered measurements, SOC/SOH, configuration limits, IMD/shutdown status	Fault flags, warnings, shutdown request to contractors/ shutdown circuit

Module	Functionality
Measurement & Filtering	Reads all sensors and AFE data, applies basic filtering/scaling, and produces clean measurement values
Balancing controller	Decides when and which cells to balance, incurring balancing only occurs in allowed conditions
CAN-FD communication	Packages internal state into CAN-FD messages from the ECU and applies configuration or commands received over CAN.
Fault Detection and Protection Logic	Compares measurements against limits, detects unsafe conditions, laches faults, and requests safe shutdown

Our system is not something the driver uses directly like an app, but it supports them through clear feedback and safe, automatic behavior. A typical driver turns on the car, and the system quietly checks all cells, and temperatures, and insulations; if everything is okay, it closes the contactors and allows the ECU to enable torque. If a fault occurs (for example, a cell overheats or voltage goes out of range), the BMS immediately opens the shutdown circuit, sends a fault message over CAN, and the driver sees a warning on the display.

3.5 Verification Plans

We're going to generate certain voltage and current values that qualify as Overvoltage, Undervoltage, and Overcurrent and the system will set the signal for the Shutdown Circuit and report each specific fault to the ECU with which specific cell(s) had the fault.

3.6 Risks

1. **Architectural Uncertainty (Centralized vs. Distributed)**
   - **The Risk:** We have not yet finalized whether to use a Centralized (simpler electronics, complex wiring) or Distributed (simpler wiring, complex communication) architecture. This decision critically affects our chip selection and PCB design. Delaying this decision could stall the entire hardware development process.
   - **Mitigation:** We will conduct a rapid trade-off analysis in the first two weeks. We will compare the wiring weight penalty of a Centralized system against the software complexity of a Distributed system and select the architecture that best fits our team's capabilities.
2. **Complexity of Active Balancing**
   - **The Risk:** We aim to research Active Balancing for higher efficiency. However, we recognize that Active Balancing is significantly more complex than Passive Balancing and carries a high risk of design failure or schedule overrun.
   - **Mitigation (Passive-First Strategy):** We will design the system with a "Passive-First" approach, ensuring the hardware natively supports Passive Balancing as a guaranteed baseline. We will only attempt to add Active Balancing features if our initial research proves it is feasible within the timeline. If not, we will proceed with the Passive-only fallback.
3. **High Voltage Safety Risk (Hardware Failure)**

- **The Risk:** Designing a system for 100V-600V requires strict adherence to creepage and clearance standards. A minor error in the PCB layout or isolation barrier can lead to high-voltage arcing, destroying the microcontroller or causing a safety hazard.

4. **Component Availability (Supply Chain)**

   - **The Risk:** Specialized BMS chips (AFEs) required for either architecture are currently prone to global shortages.

5. **Existential Risks**

   - **The Risk:** As a student team, our availability fluctuates due to exams and other coursework. Additionally, FSAE rules may undergo minor interpretations that impact design. There is a risk that we design a perfect system on paper but run out of time to debug the physical hardware or integrate it with the vehicle.
   - **Mitigation:** We will prioritize a "Minimum Viable Product" (MVP) strategy—getting the simplest version working first. Comprehensive documentation will be maintained on GitHub/Drive to ensure smooth handoff to future teams.

6. **Backup Plan (Fail-safe Strategy)**

   - **Fallback to Simplicity:** If our research into Active Balancing or Distributed Architecture proves too difficult, or if High Voltage integration becomes unsafe, our backup plan is to deliver a Low-Voltage (e.g., 48V) Passive-Balancing BMS prototype. This ensures we can still deliver a working product that meets all core Capstone requirements (Monitoring, Protection, Communication) without the risks of the full high-voltage pack.

3.7 Deliverables

Using the he Atlassian suite of tools which include Jira, Bitbucket, and Confluence to deliver on the following

- Project proposal
- Weekly Progress Reports
- Final report
- ECE Capstone Poster Session poster
- Detailed design documentation, explaining what your design does, how your design works, and why you made the decisions you did
- Electrical CAD: Schematics and board layouts, including output files (Gerbers, PDFs, etc)
- Mechanical CAD: enclosures, mechanisms, including output files (STLs, PDFs, etc)
- Bill of materials and pricing
- Short user manuals documenting how to use your creation
- A working prototype with a prototype battery pack and functionality with the Energy Control Unit (ECU) prototype.

4 Project Management Plan

4.1 Project Timeline

Here is the link to the gantt chart for our project.

4.2 Budget and Resources

Our budget as per the industry sponsor is $1000+. This should cover:

- PCB Fabrication and Assembly
- Electronic Components (cell monitoring ICs, microcontroller, sensors, actuators)
- Development/Evaluation Boards
- Testing and Miscellaneous
- Mechanical Design Enclosures

These are some of the resources that we may utilize:

- EPL
- Soldering and rework equipment
- Multimeter
- PSU labs for design and testing
- Logic analyzer
- Oscilloscope

4.3 Intellectual Property Discussion

The IP is owned by the VMS student group.

4.4 Team

4.4.1 Members and their skills

- **Hussein Khodor, Computer Engineering:** Embedded systems programming (C/C++), microcontroller development, Python scripting
- **Jose Ramirez, Computer Engineering:** Embedded systems programming (C/C++), microcontroller development, communication protocols (CAN, SPI, I2C), PCB layout and design
- **David Fransis, Electrical Engineering:** Analog circuit design, PCB layout and design, power electronics, battery chemistry
- **Fangdong Yuan, Electrical Engineering:** Analog circuit design, PCB layout and design, power electronics

4.4.2 Roles

- **Hussein Khodor, Software Team:** Data acquisition and processing
- **Jose Ramirez, Software Team:** CAN bus interfacing
- **David Fransis, Hardware Team:** Battery protection circuitry, voltage and temperature sensors
- **Fangdong Yuan, Hardware Team:** Cell balancing, fault logging

Jose Ramirez shall take the responsibility of team leader.

4.5 Development Tools and Process

The team plans to follow an iterative and agile-inspired development process, emphasizing early prototypes, incremental feature development, and frequent testing. Rather than attempting a single fully featured implementation, the project will progress through multiple functional prototypes, each adding capability and validation.

Collaboration and project management tools include:

- **Version Control:** Bitbucket (schematics, PCB layouts, firmware documentation)
- **Task Management:** Jiro for sprint planning and tracking
- **Communication:** Discord/Slack for daily communication
- **Documentation:** Confluence/Google Drive for shared documents

From a technical perspective, development tools are expected to include:

- **PCB Design:** KiCad
- **Circuit Simulator:** LTspice or KiCad simulation tools
- **Firmware Development:** Arduino IDE or PlatformIO with VSCode
- **Programming Languages:** C/C++ for embedded firmware, Python for test scripts
- **CAN Bus Analysis:** Saleae Logic 2, Sigrok Pulseview
- **3D Modeling CAD:** Fusion 360, SolidWorks, Onshape
- **Documentation:** LaTeX, Microsoft Word, Google Docs

5 Signatures

Here is the proof that our industry sponsor approved of our Proposal.

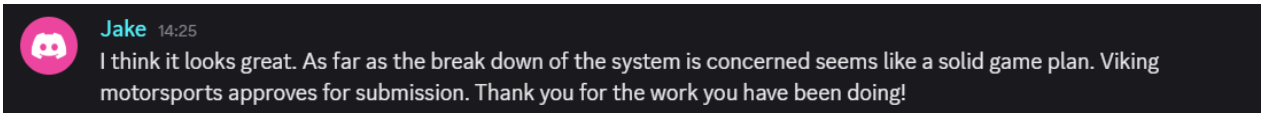


Figure 3: Sponsor Sign Off

We weren't able to get the faculty advisor signature as they are out of the country and unavailable the week of submission.